

Dataset-Free Inverse Design of Electrostatic MEMS via the Pull-In Bifurcation

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Abstract—Pull-in caps the stable travel of an electrostatic MEMS actuator at a third of its gap and destroys the device when crossed. It is a saddle-node bifurcation, where the tangent stiffness is singular and a differentiated forward solve breaks down, which is why the field designs around it with surrogates fitted to large FEM campaigns. We solve the extended fold system directly, making the bifurcation an ordinary root of a regular system so that the implicit function theorem gives exact derivatives of pull-in voltage with respect to electrode geometry. Three capabilities follow, none using a training set: inverse design, cutting drive voltage $22.86 \rightarrow 13.60$ V (−41%) at the same travel specification with no process change; an amortized network trained through the solver, emitting a geometry in 0.04 ms to within 0.56% of specification; and a control policy that infers each device’s unmeasurable fabrication limit, recovering 12.8% more usable travel with no device destroyed. We validate against eight independent sources spanning 1967–2025, reproducing a reversal of the optimal gap exponent between geometries.

I. Problem

Electrostatic actuation dominates commercial MEMS, and pull-in bounds what it can deliver. Past a critical voltage the electrostatic attraction outruns the elastic restoring force and the structure collapses onto the electrode.

Two costs follow. Staying safely below pull-in takes tens of volts, so the die carries a charge pump that spends area and power. And travel ends at a third of the gap, at a threshold that shifts with film thickness, residual stress and etch bias. Measuring that threshold requires crossing it, which destroys the device, so drive schedules are set for the worst die and the rest of the wafer is under-driven.

Design practice meets this with FEM sweeps over hand-picked geometry families, in tools that return a number and no derivative. Recent AI work inherits the handicap: the current standard pipeline [1] fits a surrogate to 48,000 simulations, then optimises through that surrogate and its approximation error. We remove the intermediary.

II. Methodology

A. Governing equations

A beam of plate modulus $\tilde{E}$ and thickness t over a shaped electrode gap $d(x)$ obeys, after nondimensionalisation with $x = l\xi$, $y = d_0Y$ and $G = D - Y$,

$$Y'''' - N_t Y'' = \Lambda \frac{1 + \alpha G}{G^2}, \quad (1)$$

where $\Lambda = 6\epsilon_0 V^2 l^4 / (\tilde{E} t^3 d_0^3)$ is the normalised electrostatic load, N_t carries residual-stress tension, and α is a fringing-field correction. Fourth-order finite differences reduce this to a residual $R(Y, \Lambda)$ on the interior nodes.

B. Differentiating through the fold

Pull-in is not the maximum of a smooth function. It is a turning point of the equilibrium branch, and there the tangent stiffness $J = \partial R / \partial Y$ is singular. A differentiated forward solve fails at precisely that point, because the linear solve inside it is the object going singular.

So we solve for the fold instead of approaching it. Following Keller [6], the equilibrium residual is extended with a null-vector condition and a normalisation,

$$R(Y, \Lambda) = 0, \quad J(Y, \Lambda) v = 0, \quad v^\top v - 1 = 0, \quad (2)$$

and the three are solved together for (Y, Λ, v) in a single Newton solve, warm started from a coarse displacement-controlled sweep. The bifurcation is now an ordinary root of a regular system, so the implicit function theorem applies and returns $\partial \Lambda / \partial d(x)$ exactly, with no finite differencing and no fitted model in between.

Being correct is not the same as converging, and two globalisation choices, both found by watching a failure, let the solver reach profiles it has never seen. The first is row equilibration: without it an $O(1)$ constraint row sits alongside beam rows of order 10^6 and Newton stalls on shaped profiles. The second is a gap floor of 10^{-12} rather than 10^{-6} , which makes the residual of a beam that has passed through the electrode enormous, so that state is rejected instead of converging onto a nonphysical branch. The stencil is cancellation-limited, so the solver runs in float64 and is CPU-bound.

III. Datasets

The exact gradient of Sec. II-B removes the need for a dataset, so this section is short by construction. Every gradient that trains every network here comes from a live solve of (2); the network of Sec. IV consumed 6,000 live solves and stored none of them. Two kinds of data do appear and we keep them apart: the surrogate baselines of Sec. V-C consume 400 sampled solves each, generated on demand, while validation targets are literature values cited at the point of use in the source file.

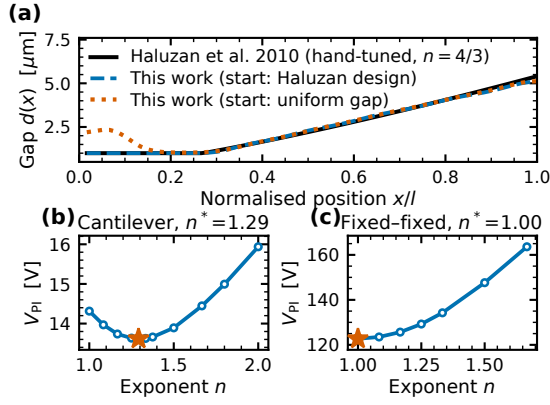


Fig. 1. (a) The optimiser rediscovers the profile of [3] from a uniform-gap start. (b), (c) The optimal exponent reverses between geometries.

IV. Model architecture

A. Amortized design network

The gap profile is parameterised as $d(\xi) = \text{softplus}(\sum_{k < K} p_k \cos k\pi\xi)$ over $K = 12$ cosine modes on $N = 40$ nodes, plus a scalar scale σ . The network maps the required travel T onto that parameterisation: two inputs, T normalised to its range and $\log T$, pass through a $2 \rightarrow 64 \rightarrow 64$ tanh trunk and split into linear heads for the K shape coefficients and for $\log \sigma$.

B. The hard-constraint layer

A soft penalty on the travel specification is biased here, and the bias has a direction. Reducing travel also reduces V_{PI} , so the optimiser gets paid for missing the specification, and it settles near 5% error. No amount of weight tuning fixes a gradient that points the wrong way.

We replace it with a differentiable enforcement layer, in the spirit of OptNet [7] and HardNet. Four Newton steps solve $\text{travel}(p, \sigma) = T$ for σ ; a fifth correction step is taken with the p -dependence live while $\partial \text{travel} / \partial \sigma$ is held fixed. At convergence that reproduces $d\sigma^*/dp = -F_p/F_\sigma$, the same implicit-function argument used for (2) itself, so the constraint holds exactly and the shape head still receives a correct gradient. The loss carries no penalty term, and the scale head is regressed onto a stop-gradient copy of the layer's answer, so inference stays one forward pass. Specification error falls from about 5% to 0.56% (Fig. 2).

C. Baselines and control policy

The strongest baseline is a 1D Fourier neural operator [8]: width 32, 12 modes, depth 4, given active learning, a trust region and feasible restarts. Our first comparison was a strawman: without a trust region the surrogate proposed designs where the true solver returned a travel of -639458 . The control policy is a $7 \rightarrow 64 \rightarrow 1$ tanh network trained by REINFORCE with a batch-mean baseline and Gaussian exploration ($\sigma = 0.10$); pathwise gradients do not apply, because episode termination is irreversible.

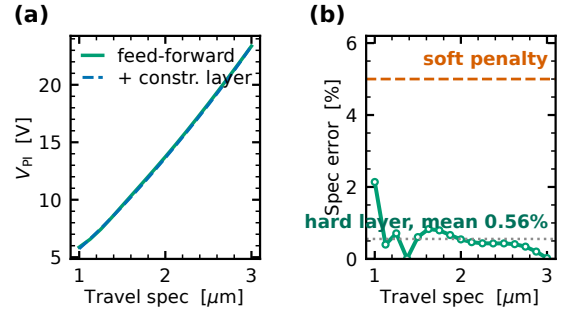


Fig. 2. The network across the whole specification range. (a) Output quality. (b) The hard-constraint layer cuts specification error from $\sim 5\%$ to 0.56%.

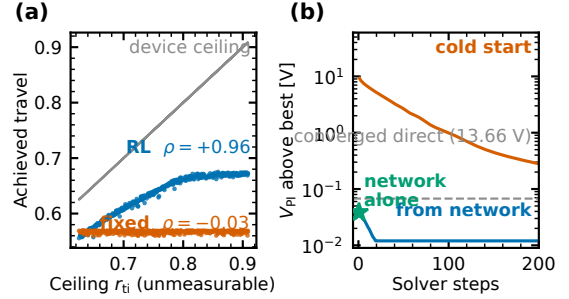


Fig. 3. (a) The policy tracks each device's unmeasurable ceiling; fixed gain cannot. (b) The network's output beats a cold start in 5 steps, not 500.

V. Results

A. Validation

Table I summarises agreement with eight independent sources published between 1967 and 2025.

The exact gradient reproduces the analytic c^3 scaling law to 0.00%. That check tests the differentiation rather than the forward solve, and is the one that would have caught a plausible but wrong adjoint.

The optimal gap exponent reverses between geometries (Fig. 1(b),(c)): $n^* = 1.29$ for the cantilever against a reported $4/3$, and $n^* = 1.00$ for the fixed-fixed beam, matching the monotone trend reported there. A model tuned to one trend does not produce the opposite trend in the other geometry by accident.

Seeger and Boser [4] supply a discriminating case: two of their designs are mechanically identical and differ only in parasitic capacitance. The model correctly predicts that this purely electrical difference changes which instability arrives first, costing 20% of travel.

B. Inverse design

Minimising V_{PI} at fixed stable travel, by gradient descent on a free-form profile with the scale set by the constraint layer so every iterate stays feasible, takes the cantilever benchmark from 22.86 V to 13.60 V. That is 41% from electrode geometry alone, with no new material, process step or mask.

TABLE I

Solver validation against eight independent sources, 1967–2025. “mech.” is the predicted instability mechanism; the last two rows are measured devices. [†]Normalised, as in [5].

Source	Quantity	Ours	Ref.	Error
Closed form	$\lambda^* = 4/27$ (lumped)	0.148150	0.148148	0.001%
Nathanson [9]	Λ_{PI} , cantilever	1.6799	1.6790	0.05%
Nathanson [9]	Λ_{PI} , fix–fix	69.935	71.167	1.73%
Analytic c^3 law	$\sum \partial\Lambda/\partial D$	5.0398	5.0398	0.00%
Seeger [4]	tip-in, design #1	19.0%	19%	—
Seeger [4]	tip-in, design #3	85.1%	85%	—
Seeger [4]	mech., 3 designs	3/3	3/3	—
Nazemi [5]	V_{PI} reduction [†]	13.2%	16.0%	2.8pp
M-TEST [2]	V_{PI} , 500 μm	11.70 V	11.1 $\pm$ 0.1 V	+5.4%

TABLE II

Inverse design versus [3], both boundary conditions, every design re-fitted to 2 μm stable travel at $N = 60$.

Design	Cantilever		Fixed–fixed	
	Ours	Paper	Ours	Paper
Uniform gap	22.86	23.66	175.83	178.26
Linear ($n=1$)	14.31	14.86	123.01	125.48
Best family in [3]	13.61	14.14	121.79	123.94
Free-form (ours)	13.60	—	120.03	—
Optimal exponent n^*	1.29	4/3	1.00	1.00
Gain over best family	0.14%	—	1.45%	—

Starting from a uniform gap and given nothing about any prior design family, the optimiser rediscovers the hand-tuned profile of [3] (Fig. 1(a)) and descends 22.9 $\rightarrow$ 13.9 V unaided, ruling out a favourable initialisation. Two quantities came out matched without ever being fitted: a maximum gap of 6.571 μm against their 6.57, and a flattened-bottom ratio of 0.80 against 0.78. Table II reports both boundary conditions.

That paper conjectured that gains beyond simple shapes “are believed to be minimal” but had no way to test it. Free-form optimisation measures what remains: 0.14% on the cantilever, 1.45% on the fixed–fixed beam.

C. Comparison with the standard pipeline

Table III places this work against how MEMS design with machine learning is currently done. Zhang et al. [1] generate 48,000 finite-element simulations, fit an MLP, and inverse-design through that surrogate, which returns a design in 0.01 s where a full FEA solve takes 22 s. The contrast is the dataset column: we store no simulations at all, because the gradient comes from the physics, and still emit a design in 0.04 ms.

The lower rows re-implement both surrogate families on our own task. At an equal budget of 400 true solves an MLP reaches 17.20 V and an FNO 14.25 V, against 13.78 V for direct differentiation and 13.64 V for the network. The FNO is competitive on quality; what separates us is needing no training data, no architecture search and 40 $\times$ less wall-clock. One forward pass lands inside the 0.28% spread across optimiser restarts.

TABLE III

Design methodologies compared. [1] values are as reported there, for a different task, so the comparison is of methodology. Rows 3–8 share one task at an equal 400-solve budget, scored by the exact solver.

	Method	Sims. in dataset	Gradient source	Time per design	V_{PI} [V]
[1]	MLP fitted to FEM data	48,000	surrogate	0.01 s	—
[3]	Hand-tuned gap family	—	none	manual	14.14
	MLP surrogate + act. learn.	400	surrogate	71 s	17.20
	FNO surrogate + act. learn.	400	surrogate	594 s	14.25
This work	Direct differentiable	0	exact	15 s	13.78
	Direct, 4 restarts	0	exact	170 s	13.67
	Amortized network	0	exact	0.04 ms	13.64
	Network + 100 steps	0	exact	17 s	13.60

D. Recovering travel lost to tolerance

The tip-in ceiling $r_{ti} = 1/(1 + k_x L_c^2/6k_\theta)$ varies device to device and cannot be observed at test time, so fixed-gain control must aim at the worst case and forfeits 19% of travel. A policy seeing only the device’s own tilt response recovers $80.4 \pm 4.9\%$ of that range, 12.8% more usable travel, over 1,536 held-out episodes with $0.0 \pm 0.0\%$ destroyed.

Reduced conservatism alone would give the same average, so the average is not the evidence. Adaptation shows up in the correlation between the travel a device achieves and its own ceiling: $\rho = +0.96$ against -0.03 for fixed gain (Fig. 3(a)). The policy tracks what it cannot measure.

VI. Conclusion

Solving the extended fold system directly turns pull-in from an obstacle into a differentiable object, and the exact gradient that follows removes the dataset from AI-driven MEMS design: against the 48,000 simulations of the standard pipeline [1] we store none, and still emit a geometry in 0.04 ms. On the benchmark of [3] this cuts drive voltage 22.86 $\rightarrow$ 13.60 V from geometry alone, and in closed loop the same physics recovers 12.8% of the travel fabrication tolerance otherwise forces a designer to surrender, with no device destroyed. Absolute voltages still carry an offset traced to anchor compliance a 1D model cannot represent; adding it is our next step.

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